L6: Congruences

- Note that Proposition 2.4 from last lecture can be stated in terms of equivalence classes under congruence modulo m :
  (a) $[a]+[c]=[a+c]$
  (b) $[a][c]=[ac]$.

This allows us to construct addition and unultiplication tables for equivalence classes under congruence modulom using any Complete residue system modulo m.

Example:	+	[0]	[1]	[2]
Modulo 4:	[0]	[0]	[1]	[2]
	[1]	[1]	[2]	[3]
	[2]	[2]	[3]	[0]
	[3]	[3]	[0]	[1]
	⋮	[0]	[1]	[2]
	[0]	[0]	[0]	[0]
	[1]	[0]	[1]	[2]
	[2]	[0]	[2]	[0]
	[3]	[0]	[3]	[2]
	[2]]			

- $\mathbb{Z}_4 := \{[0],[1],(2),[3]\}$ along with the operations of addition and multiplication form a ring.

- Consider the multiplication and addition tables for equivalence classes under congruence modulo 5:

+	[0]	[1]	[2]	[3]	[4]
[0]	[0]	[1]	[2]	[3]	[4]
[1]	[1]	[2]	[3]	[4]	[0]
[2]	[2]	[3]	[4]	[0]	[1]
[3]	[3]	[4]	[0]	[1]	[2]
[4]	[4]	[0]	[1]	[2]	[3]
[0]	[0]	[1]	[2]	[3]	[4]
[0]	[0]	[0]	[0]	[0]	
[1]	[0]	[1]	[2]	[3]	[4]
[2]	[0]	[2]	[4]	[1]	[3]
[3]	[0]	[3]	[1]	[4]	[2]
[4]	[0]	[4]	[3]	[2]	[1]

- $\mathbb{Z}_5 := \{[0],[1],[2],[3],[4]\}$ along with the operations of addition and multiplication form a field.

Question: Why does $\mathbb{Z}_5$ have more structure than $\mathbb{Z}_4$?

- Cancellation law for multiplication does not hold in general for congruence modulo m.
  Proposition 2.5:
  Let $a, b, c \in \mathbb{Z}$. Then $ca \equiv cb(\bmod m)$ if and only if $a \equiv b \bmod (m/(c,m))$.
  Proof:
  $\Leftrightarrow$ direction): Assume that $ca \equiv cb(\mathrm{mod} m)$ and let $d := (c, m)$. We have $m \mid ca - cb = c(a-b)$ and thus m/d $\mid (c/d)(a-b)$, and so $a \equiv b(\bmod \text{ m/d})$ Since (m/d, c/d) = 1 by Proposition 1.10.
  ($\Leftarrow$ direction) : Let $d := (c, m)$ and assume that $a \equiv b(\mathrm{mod} m/d)$. Then $m/d \mid a-b$, so that implies that $m \mid da-db$. So $m \mid (4d)(da-db) = ca-cb$ and $ca \equiv cb(\mathrm{mod} m)$, as desired.

Linear congruences in one variable:

Definition: Let $a, b \in \mathbb{Z}$. A congruence of the
form $ax \equiv b(\mathrm{mod} m)$ is said to be a linear congrnence in the variable x.
Example: $2x \equiv 3(\mathrm{mod} 4)$ has no solutions in $\mathbb{Z}$. Check $x \equiv 0, 1, 2, 3(\mathrm{mod} 4)$ don't work.

- $2x \equiv 3(\mathrm{mod} 5)$ has one solution modulo 5 , namely $x \equiv 4(\mathrm{mod} 5)$, and so has solution Set in $\mathbb{Z}$ given by [4].

- $2x \equiv 4(\mathrm{mod} b)$ has two solutions modulo b, namely $x \equiv 2, 5(\mathrm{mod} 6)$, and so has solution Set in $\mathbb{Z}$ given by [2] U[5].

- $3x \equiv 9(\mathrm{mod} 6)$ has three solutions modulo 6) namely $x \equiv 1, 3, 5(\mathrm{mod} 6)$, and so has solution set in $\mathbb{Z}$ given by $[1] \cup [3] \cup [5]$.

Theorem 2.6: Let $ax \equiv b(\mathrm{modm})$,
and let $d = (a, m)$. If $d + b$, then the congruence has no solutions in $\mathbb{Z}$. If $d \mid b$, then the congruence has exactly d incongruent solutions modulo m in $\mathbb{Z}$.
Proof: Observe that $ax \equiv b \bmod m)$ if and only if $m/ax - b$ if and only if $ax - b = my$ for some $y \in \mathbb{Z}$ if and only if $ax - my = b$ for some $y \in \mathbb{Z}$. Thus $ax \equiv b(\mathrm{mod} m)$ is Solvable in x if and only if $ax - my = b$ is solvable for x and $y(*)$.
Since $d \mid a$ and $d \mid m$, we have that $d \mid ax - my$ by Proposition 1.2, which implies that $d \| b$. Equivalently, if $d + b$, then $ax \equiv b(\mathrm{mod} m)$ has no solutions. Wlog now assume that dlb.
We now prove the Second part of the Thervem in 4 steps:
(a) We show that $ax \equiv b(\mathrm{mod} m)$ has a solution, say x_0.
(b) Given any solution x_0, we show that
$ax \equiv b(\mathrm{mod} m)$ has infinitely many solutions in $\mathbb{Z}$ of a given form.
(c) Given any solution x_0, we show that every solution to $ax \equiv b(\mathrm{mod} m)$ has the form in (b) (from which (b) and (c) combine to yield all solutions in $\mathbb{Z}$ to $ax \equiv b(\mathrm{mod} m)$).
(d) We show that there are exactly d incongruent solutions modulo m in $\mathbb{Z}$.

For (a): By Proposition 1.11, there exist $r, s \in \mathbb{Z}$ such that $d = (a, m) = ar+ms$. Furthermore, $d1b$ implies that $b = de$ for some $e \in \mathbb{Z}$. Then

$$b = de = (ar + ms)e = a(re) + m(se).$$

Thus $x =$ re and $y = -$ se solve $ax - my = b$,
from which $x_0 =$ re solves $ax \equiv b$ (modm)
by (*).

For (b): Let x_0 be any solution of $ax \equiv b (\text{mod} m)$. Let $n \in \mathbb{Z}$ and consider $x_0 + \left(\frac{m}{d}\right) n$. Note that $x_0 + \left(\frac{m}{d}\right) n$ is an integer, since d/m. Furthermore,

$$\begin{aligned} a\left(x_0 + \left(\frac{m}{d}\right) n\right) &= ax_0 + a\left(\frac{m}{d}\right) n \\ &= ax_0 + \left(\frac{a}{d}\right) mn \\ &= b + 0(\text{mod} m) \\ &\equiv b(\text{mod} m). \end{aligned}$$

Thus given any solution x_0 to $ax \equiv b(\text{mod} m)$, the infinitely many values $x_0 + \left(\frac{y}{d}\right) n$ in $\mathbb{Z}$, $n \in \mathbb{Z}$, are also solutions.
For (c) : Let x_0 be any solution to $ax \equiv b(\text{mod} m)$. Then $ax_0 - my_0 = b$ for some $y_0 \in \mathbb{Z}$ by
Any other solution x of $ax \equiv b$ (modm) implies the existence of $y \in \mathbb{Z}$ such that $ax - my = b$ by (*), and so we have

$$(ax - my) - (ax_0 - my_0) = b - b = 0.$$

Thus $a\,(x - x_0) = m\,(y - y_0)$.

Dividing both sides by $d := (a, m)$ we obtain $\frac{a}{d}\,(x - x_0) = \frac{m}{d}\,(y - y_0)$.
Now $\frac{m}{d} \,\Big|\, \left(\frac{a}{d}\right)(x - x_0)$; since $\left(\frac{a}{d}, \frac{m}{d}\right) = 1$ by Proposition 1.10, we have that $\frac{m}{d}\Big|\ x - x_0$. Consequently $x - x_0 = \left(\frac{m}{d}\right)_n$ for some $n \in \mathbb{Z}$,
So equivalently, $x = x_0 + \left(\frac{m}{d}\right) n$ for some $n \in \mathbb{Z}$. Hence, given a solution x_0 of $ax \equiv b$(modm), we have that every solution takes the form $x_0 + \left(\frac{m}{d}\right) n$ with $n \in \mathbb{Z}$. Combining this vesult with part (b), we have that all solutions are given precisely by $x_0 + \left(\frac{y}{d}\right)_n$ with $n \in \mathbb{Z}$.
For (d): We find necessary and sufficient conditions for the congruence modulo m of two solutions in the family $x_0 + \left(\frac{m}{d}\right) n, n \in \mathbb{Z}$.

We have

$$x_0 + \left(\frac{m}{d}\right) n_1 \equiv x_0 + \left(\frac{m}{d}\right) n_2 (\text{mod} m)$$

if and only if

$$\left(\frac{m}{d}\right) n_1 \equiv \left(\frac{m}{d}\right) n_2 (\text{mod} m)$$

if and only if

$$m \,\Big|\, \left(\frac{m}{d}\right)(n_1 - n_2)$$

if and only if (by Proposition 2.5)

$$d \mid n_1 - n_2$$

if and only if

$$n_1 \equiv n_2 (\bmod d).$$

Thus two solutions of the form $x_0 + \left(\frac{m}{d}\right) n$, $n \in \mathbb{Z}$, are congruent modulo m if and only if the corresponding n-values are congrwent modulo d. Thus a complete set of incongrnent Solutions modulo m of $ax \equiv b(\bmod m)$ is obtained from $x_0 + \left(\frac{m}{d}\right)^n, n \in \mathbb{Z}$, by letting n range over a complete residue system modulo d, say $\{0, 1, \ldots, d-1\}$ for example.

Corollary 2.7: Consider $ax \equiv b(\bmod m)$
and let $d = (a, m)$. If $d \| b$, then there are d incongruent solutions modulo m, given by

$$x_0 + \left(\frac{m}{d}\right) n, \quad n = 0, 1, 2, \ldots, d-1,$$

where x_0 is any particular solution to the Congruence.
Example: Find all incongruent solutions to $16x \equiv 8(\bmod 28)$ Lets use the Euclidean algorithm to Compute

$$\begin{aligned} d = (16, 28); 28 &= (6.1 + 12 \\ 16 &= 12.1 + 4 \\ 12 &= 4.3 + 0, \end{aligned}$$

So $d = 4$. Since 418 , the Congruence has four incongruent solutions modulo 28 by
Theorem 2.6. Working backward through the
Euclidean algorithm we obtain

$$4 = 2 \cdot 16 + (-1)28.$$

Thus $8 = 4 \cdot 16 + (-2) \cdot 28$,
and so $x_0 = 4$ is a particular solution to $16x \equiv 8(\bmod 28)$. Thus by Corollary 2.7, all incongruent solutions to $16x \equiv 8(\bmod 28)$ are given by

$$4 + \left(\frac{28}{4}\right) n, n = 0, 1, 2, 3,$$

or equivalently, $x \equiv 4, 11, 18, 25(\bmod 28)$.
Definition: Any solution of $ax \equiv 1(\bmod m)$ is said to be the multiplicative inverse of a modulo m.

Corollary 2.8 (to Theovem 2.6): The Congruence $ax \equiv 1(\bmod m)$ has a solution if and only if $(a, m) = 1$.
If $(a, m) = 1$, then $ax \equiv 1(\bmod m)$ has exactly one incongruent solution modulo m.
Example: Find all incongruent solutions to $5x \equiv 12(\bmod 16)$. Since $(5, 16) = 1$,

the congmence has a unique solution modulo 16 by Theovem 2.6. Use the method of the
previous example (exercise) to show that 13 is the inverse of 5 modulo 16. Thus

$$13 \cdot 5x \equiv 13 \cdot 12 (\mathrm{mod} 16)$$

and so

$$\begin{aligned} x &\equiv 12.13 (\mathrm{mod} 16) \\ &\equiv (-4)(-3) (\mathrm{mod} 16) \\ &\equiv 12 (\mathrm{mod} 16). \end{aligned}$$